Docker commands

Container commands

Command	Description
docker run <image>	Create and run a new container
docker run -p 8080:80 <image>	Publish container port 80 to host port 8080
docker run -d <image>	Run a container in the background (detached)
docker run -v <host>:<path> <image>	Mount a host directory to a path in the container
docker run -v <host>:<path>:ro <image>	Mount the directory as a readonly volume
docker ps	List running containers
docker ps -a	List all containers (running or stopped)
docker stats	Display a live stream of resource usage statistics
docker logs -f <container_name>	Follow the log stream of a container
docker stop <container_name>	Stop a running container
docker start <container_name>	Start a stopped container
docker rm <container_name>	Remove a container

Executing commands in a container

Command	Description
docker exec <container_name> <command>	Execute a command in a running container
docker exec -it <container_name> bash	Open an interactive shell in a running container

Image commands

Command	Description
docker build -t <image> .	Build and tag a new image from a Dockerfile
docker images	List local images
docker rmi <image>	Remove an image
docker image prune	Remove all unused images

DOCKER HUB

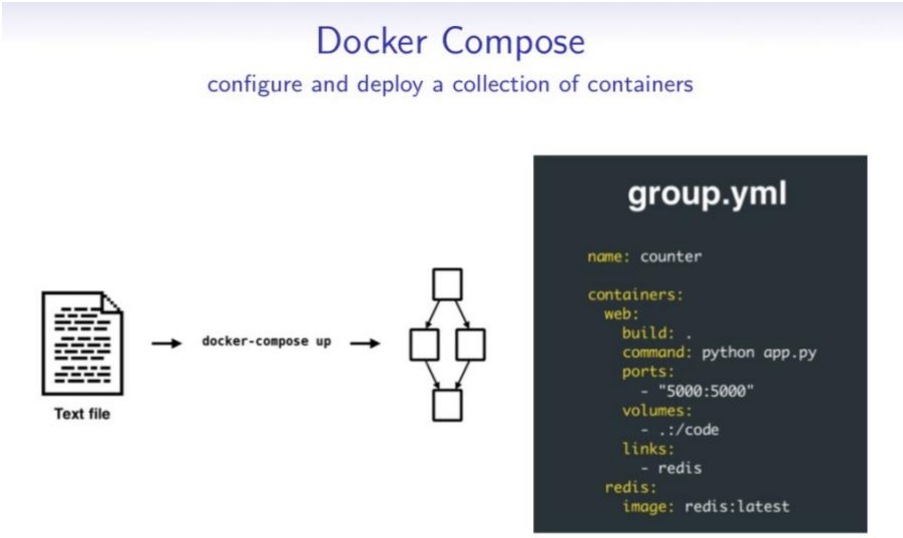
Docker Hub is a service provided by Docker for finding and sharing container images with your team. Learn more and find images at https://hub.docker.com

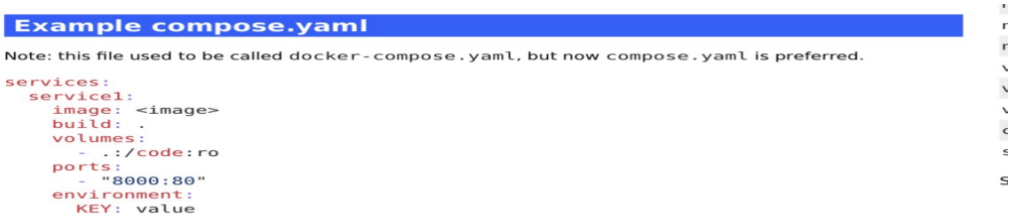
Login into Docker
```
docker login -u <username>
```

Publish an image to Docker Hub
```
docker push <username>/<image_name>
```

Search Hub for an image
```
docker search <image_name>
```

Pull an image from a Docker Hub
```
docker pull <image_name>
```





Containers commands

1- docker run -it image_name
2- Inside the image's container commands you can run
   Creating new directory- mkdir coders
   List of files inside the container- la
   Environment variables- env
   Stopping the container- exit
3- Containers exists- docker ps -a
4- Running containers – docker ps
5- Docker rm container_name – to delete or destroy the container
6- Versions of images
   Docker pull image:version
7- Docker run-d image_name
8- Renaming a container
   Docker run - - name cont_name -d image name

Additional

Docker run -d -e MYSQL_ROOT_PASSWORD=secret - - name mysql-older mysql:8.0

9- Port binding – docker run -p8080:3306 image_name
10- Trouble shoot commands
11- Docker log cont_id
12- Docker exec -it cont_id /bin/bash or /bin/sh
13-